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Customer Insights and Consumption Behavior Analysis

Funky's Hot Sauce Factory

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March, 2026

Introduction

This project investigates the spending and consumption behavior of online customers of Funky's Hot Sauce Factory, a hot sauce business headquartered in Bellingham, WA, with a customer base that spans North America. The dataset analyzed in this project contains relevant figures including basic customer information (name, customer ID), amount spent (USD), number of orders, geographic location (city, state/province, and country), marketing acceptance, and a binary indicator of tax-exempt status.

The objective of this analysis is to identify patterns and key performance indicators (KPIs) in customer purchasing behavior, evaluate the relationship between marketing engagement and revenue generation, and reveal geographic trends that may inform strategic business decisions.

Summary

This analysis of reveals clear patterns in spending behavior, marketing engagement, and geographic concentration that have meaningful implications for business strategy.

- Marketing subscribers spend significantly more than non-subscribers.
- Customer spending is highly concentrated in high-value spending tiers.
- A large portion of the customer base have yet to make a purchase.
- Revenue is geographically concentrated in Washington, U.S..

The findings in this analysis suggest that marketing engagement plays a critical role in revenue generation and that opportunities exist to expand geographically and increase customer spending.

Methodology

Analysis Tools and Technology

Data management and manipulation were performed in MySQL, a relational database management system based on structured query language (SQL). The statistical software used for data visualization and statistical analysis was R. In addition to R, Google Sheets was also used to generate data visuals.

Data Collection

The dataset for this project was collected from the company's Shopify, which serves as their online storefront. A CSV of raw customer data from all customers was downloaded from Shopify and loaded into a table in MySQL with the same columns. A cleaned customer data table was then created from this raw table by removing irrelevant fields and retaining only features relevant to the analysis. This table will be referred to as "Customers."

The features, or variables, identified for the analysis and included in the cleaned table are:

- **customer_id** (integer) – Unique identifier for each customer
- **first_name** (string) – Customer's first name
- **last_name** (string) – Customer's last name
- **email_opt_in** (string: "yes" / "no") – Whether the customer opted in for marketing emails
- **city** (string) – Customer's city
- **province** (string) – Customer's province or state
- **country** (string) – Customer's country
- **sms_opt_in** (string: "yes" / "no") – Whether the customer opted in for SMS marketing
- **total_spent** (decimal) – Total amount spent by the customer (USD)
- **total_orders** (integer) – Total number of orders placed

Data Considerations and Exploration

The total number of customers (**C**) in the recorded sample was 3,336, which was then split into:

- 1,933 (57.94%) Spending Customers (**SC**) - Customers who made at least one purchase and spent more than the minimum amount of $0.40 (minimum amount set as highest possible discount, 20%, on cheapest item, $0.50)
- 1,381 (41.4%) Non-Spending Customers (**NSC**) - Customers who had yet to make any purchases.
- 22 (0.66%) Irregular Customers (**IC**) - Customers who had orders with irregularly small or $0.00 transactions on their end.

The analysis excluded IC, allowing clearer insight on revenue-generating behavior and preventing distortion of spending metrics by customers with non-standard transactions.

SC Descriptive Statistics and Distribution

	Obs.	s	$\bar{x}$	Max	Min	1st Quartile	Median	3rd Quartile
Total Spent ($118,269.40)	1933	$56.30	$61.18	$695.25	$0.50	$33.00	$48.00	$68.00
Total Orders (2,603)	1933	0.98	1	15	1	1	1	1

Table 1: Descriptive statistics (for SC)

Table 1 summarizes the distribution of total spending and total orders for the 1,933 SC. For total spending, the mean ($61.18) exceeds the median ($48.00), indicating a right-skewed distribution. This is further supported by the maximum spent value of $695.25 relative to the quartiles. Specifically, 25% of customers spent $33.00 or less, while 75% spent $68.00 or less.

Using the inner quartile range (IQR) of 35, a reasonable spending fence can be set as [$0.00, $120.50] from calculating $[1^{st}$ Quartile $-1.5(\text{IQR}), 3^{rd}$ Quartile $+1.5(IQR)]$. Using SQL to filter out customers who spent outside of this range, 175 (9.05% of SC) customers were found to be high-value outliers, with spending values over $120.50.

In contrast, the distribution of total orders shows that most customers place very few orders. The first quartile, median, and third quartile are all equal to 1, revealing that at least 75% of customers placed only a single order. However, the maximum value of 15 orders suggests that a small subset of customers are consistent repeat buyers.

To get a visual overview of the distribution of both online spending and order values at Funky's Hot Sauce Company, histograms of total spent and total order values of SC were generated.

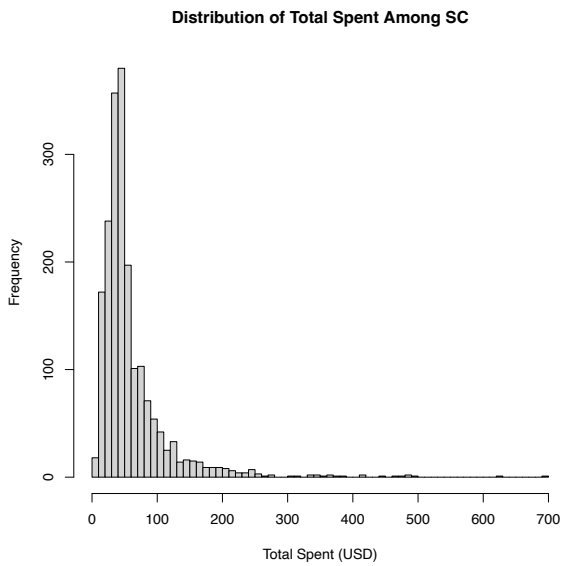

Figure 1: Spending distribution

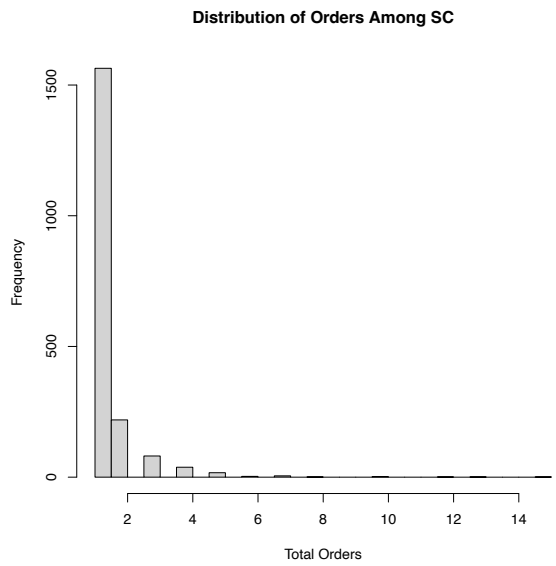

Figure 2: Order distribution

Both Figure 1 (spending) and Figure 2 (orders) portray right-skewed distributions, confirming the effect of high-value spending and order outliers on the data.

The Shapiro-Wilk normality test is useful here in determining whether spending is normally distributed (Mishra et al.). This was done in R using the built-in function `shapiro.test(x)`. Inputting total spent as the vector of interest (x), `shapiro.test(spenders$total_spent)`, a p-value of less than 2.2e-16 is returned, much less than a significance value of 0.05. Therefore, overall online spending at the company is not normally distributed.

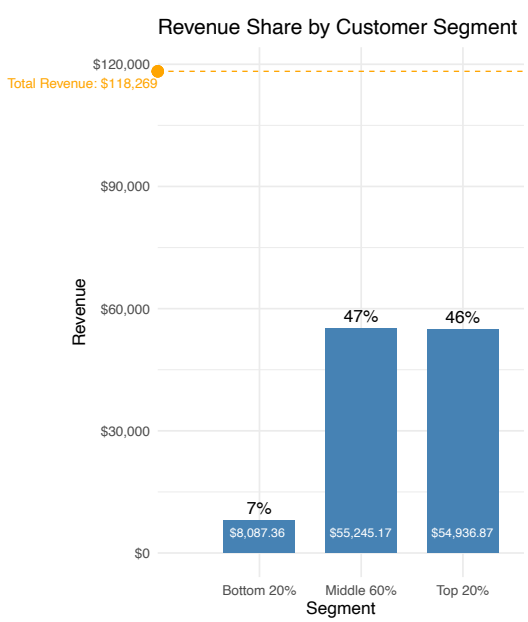

Figure 3: Segmented spending bar graph

In segmenting SC into the top 20%, middle 60%, and bottom 20% of spenders based on total spending per group compared to Funky's Hot Sauce Factory's total revenue from online sales revenue, a bar graph was created to display a visual representation of revenue share (see Figure 3).

Approximately 47%, almost half of total revenue from spending, came from the middle 60% of SC. The top 20% of spenders were closely behind, with spending making up about 46%. This aligns with the earlier findings on high-value outlying spenders.

Not represented in Figure 3 is the average amount spent among each group. The top 20% of SC total spending amounts averaged at $141.96, the middle 60% had an average of $47.63, and the bottom 20% averaged $20.95.

A closer look at spending segments gives a better illustration of how revenue share is distributed. Figure 4 displays a Pareto chart that conveys revenue contributions among SC. The bars represent spending tiers in descending order from left to right, with Tier 1 corresponding to the top 20% of spenders and Tier 5 representing the bottom 20%. Here, the middle 60% of SC is further broken up into three segments, allowing for clearer insight on which groups are vital in revenue generation.

The cumulative percentage line (red) indicates that the top 20% of customers make up 46.4% of total SC revenue, and the top 60% generate 81.4% of overall revenue from customer spending.

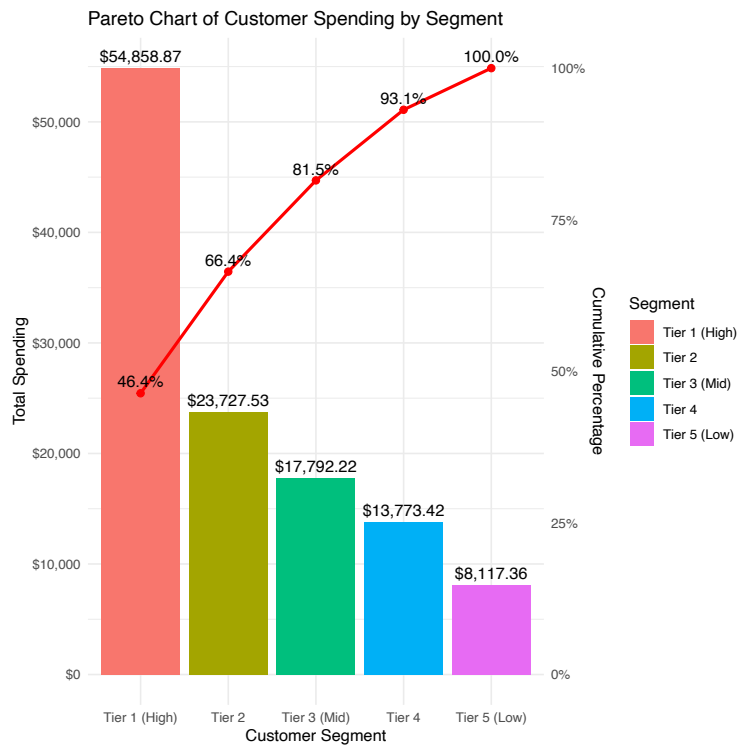

Figure 4: Pareto chart for segmented spending

Marketing Acceptance Analysis

The reasonable assumption was made that marketing subscription acceptance (email/SMS) is an advantageous insight into customer spending behavior. To explore this, a possible statistical tool is a t-test, which assesses whether there is a significant difference in the means (averages) of two independent samples—here, the difference in average spending between subscribed and non-subscribed SC. However, this test relies on both samples coming from normal distributions.

The Shapiro-Wilk normality test was first used to determine if spending amounts among subscribed and non-subscribed SC were independently normally distributed. Both groups' tests returned p-values small enough to reject normality.

Due to the non-normality of these groups' spending distributions, a non-parametric test for statistical significance in spending differences between the two was necessary. The Wilcoxon signed-rank test is exactly this. The null hypothesis here was that non-subscribed SC spending, n, was identically distributed to that of subscribed SC, s (no difference, $\Delta = 0$). The alternative hypothesis was that there is a positive difference in the distributions. Specifically, subscribed SC spend a significant amount more than non-subscribed SC ($\Delta > 0$).

$$H_0 : n \stackrel{d}{=} s + \Delta, \Delta = 0$$

$$H_1 : n \stackrel{d}{=} s + \Delta, \Delta > 0$$

Using the built-in R function `wilcox.test(s$total_spent, n$total_spent, alternative = "greater")`, the resulting p-value was 7.898×10^{-12}. Since this p-value is less than the significance level of 0.05, the null hypothesis is rejected. Therefore, there is sufficient evidence to conclude that subscribed SC spend significantly more than non-subscribed SC ($n \stackrel{d}{=} s + \Delta, \Delta > 0$).

This test is often interpreted as a test for a difference in central tendency between the two groups (Nahm). Figure 5 displays the smoothed spending distributions for both subscribed and non-subscribed SC, illustrating that the distribution for subscribed SC is shifted to the right. In particular, the median spending for subscribed SC is \$56.00, compared to \$46.00 for non-subscribed SC, further supporting this conclusion.

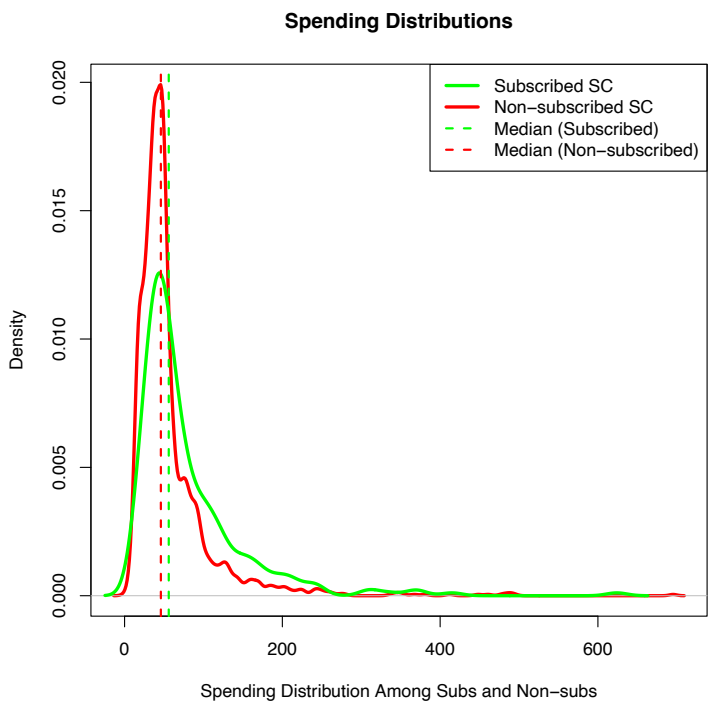

Figure 5: Subscribed SC vs. non-subscribed SC smoothed spending distribution

At the time of this analysis, 275 SC were subscribed to email and/or SMS marketing, 14.23% of all SC and 8.24% of the total customer sample, C. Given the findings from the Wilcoxon signed-rank test, a closer look into average spending amounts revealed that on average, subscribed SC spent about 39.2% more than non-subscribed SC at \$80.59 average total spending (s) versus \$57.97 (n). This notable difference suggests that marketing subscription status may be a key performance indicator of higher customer spending behavior, highlighting its potential value for targeted marketing.

In addition to this, as mentioned earlier in this report, there were also 1,381 non-spending customers, NSC, which make up 41.4% of the total customer sample. 102, 7.39% of the NSC were subscribed to some type of marketing, revealing an opportunity to potentially convert these customers into active spenders through effective marketing strategies, ultimately increasing overall customer engagement and revenue.

Geographic Insights

Geographic variation in spending behavior was also explored with the goal of assessing how customer engagement differed across regions. This offers additional context for identifying high-performing areas and potential opportunities for improvement in geographic marketing.

Most sales were from the U.S., though there was some spending data across three provinces in Canada: British Columbia, Alberta, and Ontario. Canada accounted for 0.63% of online sales revenue and the U.S. made up the rest (99.37%). This geographic analysis primarily focused on the U.S., but to get a picture of Candadian sales, Table 2 displays data from spending behavior in Canada, including the individual provinces mentioned above.

Region	Orders	Avg. Per Customer ($USD)	# of SC	Sales Revenue	Country Rank
Canada	11	$92.89	8	$743.08	N/A
BC, CA	5	$63.416	3	$317.08	1
AB, CA	3	$86.33	2	$259.00	2
ON, CA	3	$55.67	3	$167.00	3

Table 2: Canada online sales

Spending Across the U.S.

As discussed, the U.S. accounts for nearly all revenue share from online sales at Funky's. Spending is scattered across the U.S. but there are a couple high-performing states that are of note.

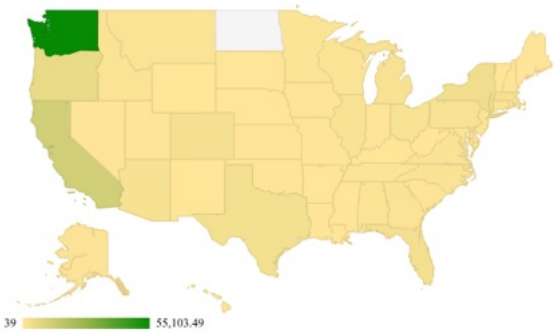

Figure 6: Density map of U.S. sales

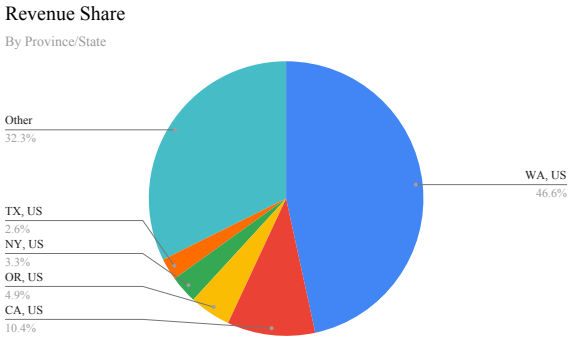

Figure 7: Revenue share by U.S. state

Figure 6 represents a density map of online sales revenue for each state in the U.S., with Washington state appearing darkest at $55,103.49. The lowest performing state was Rhode Island, with only $39.00 in sales. A closer look at the numbers revealed the top five states in sales revenue generation: Washington, California, Oregon, New York, and Texas. Figure 7 highlights these states' revenue share visually, and Table 3 provides the corresponding exact figures for each state. Washington makes up almost half, 46.6% of all online sales revenue, whereas the other four high-performing states share less than 25%, indicating a concentration of revenue generation in Washington relative to other regions.

Region	Orders	Avg. Per Customer ($USD)	# of SC	Sales Revenue	Revenue Share
WA, U.S.	1154	$62.19	886	$55,103.49	46.59%
CA, U.S.	287	$66.56	184	$12,246.26	10.35%
OR, U.S.	120	$62.65	92	$5,763.45	4.87%
NY, U.S.	96	$71.5	54	$3,861.05	3.26%
TX, U.S.	70	$55.64	55	$3,060.3	2.59%

Table 3: U.S. top five states' online sales

In addition to this, the top five cities in the U.S., based on online sales revenue and regardless of state rank, were Bellingham, WA, Seattle, WA, Portland, OR, Anacortes, WA, and Everett, WA. Four of these cities are in Washington, further supporting a revenue concentration in Washington.

Focusing again on the highest-performing states in the U.S. by online sales revenue generation, the top two cities within each state were analyzed to gain deeper insight into local spending variation and highlight the primary contributors to revenue. Figure 8 includes a bar graph comparing these cities, presenting clear visualization of revenue differences.

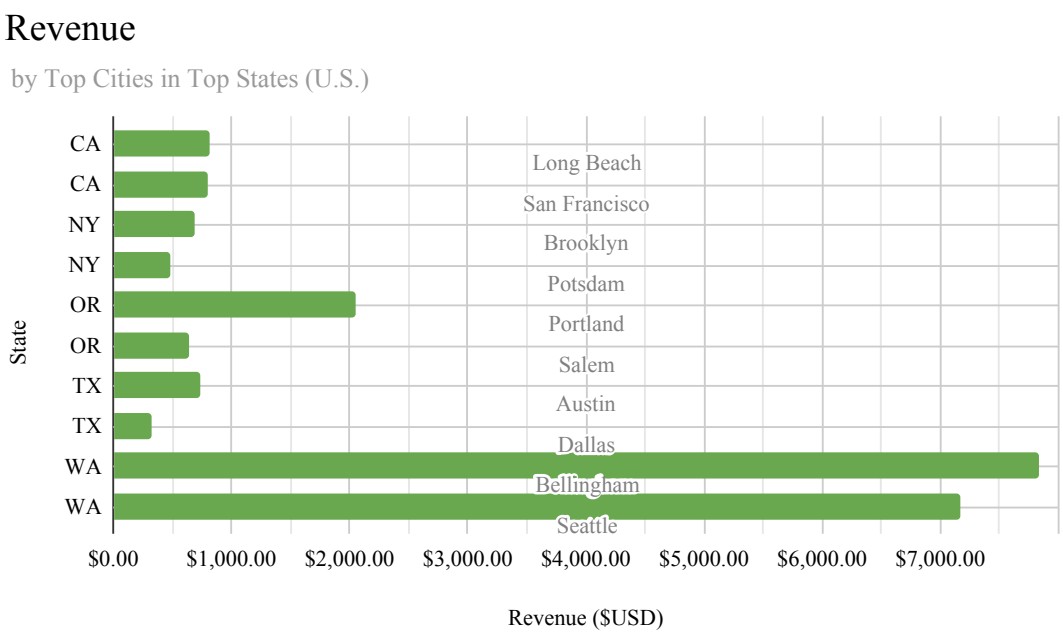

Figure 8: Bar graph of top cities in the U.S. by online sales revenue generation

U.S. Geographic Subscription Analysis

It was found that subscribed SC spent a statistically significant amount more than non-subscribed SC. Revisiting this from a geographical perspective, the proportion of subscribed SC in each city (both Canada and the U.S. included) was compared to the median spent

— using the median due to the non-normality of the data — from each respective region and then graphed alongside all other cities to visualize the differences. Figure 9 represents a scatterplot of all cities in Canada and the U.S. that had online spending data.

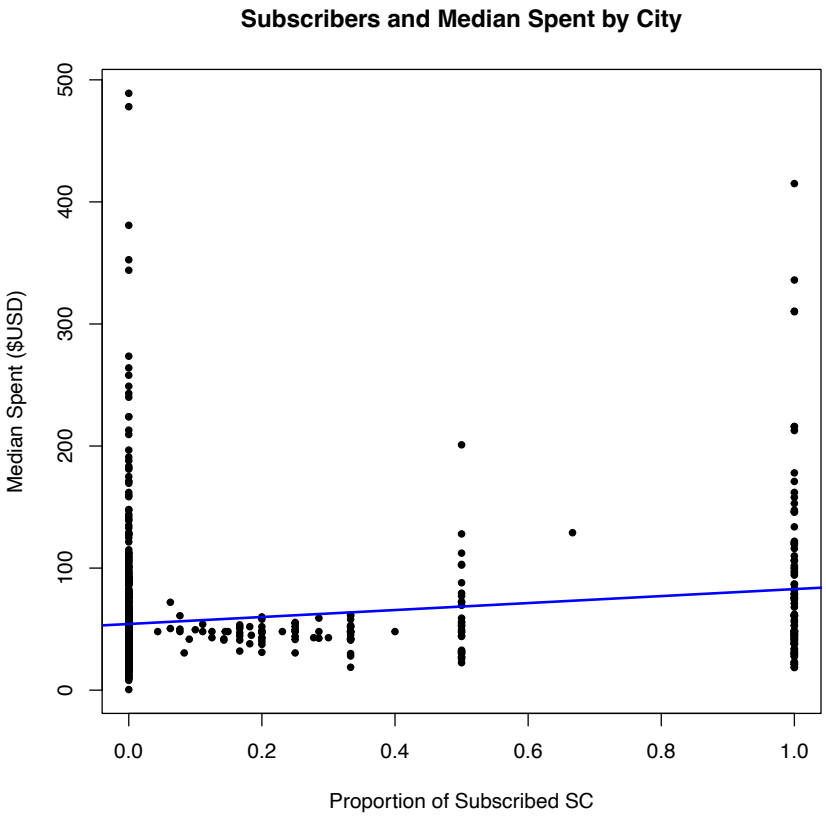

Figure 9: Scatterplot of proportion of subscribed SC against median spent by city

As displayed in the figure above, as the proportion of subscribers increases, the median amount spent also appears to increase, consistent with the fitted linear regression model that was generated in R, represented by the blue line. A test of significance for the relationship between the two variables — proportion of subscribed SC and median spent — yielded a p-value of less than 0.05, suggesting a statistically significant relationship between the two. This reinforces the earlier observed results in this analysis on the correlation between marketing acceptance and higher spending among subscribed SC.

Key Findings

This analysis identified several patterns in customer behavior and revenue distribution. Marketing-subscribed spending customers were found to spend approximately 39% more on average than non-subscribers, with statistical testing confirming a significant difference in spending behavior between the two groups. Overall spending was found to be right-skewed, with a relatively small group of high-value customers contributing disproportionately to total revenue. In particular, the top 20% of customers generated 46% of online sales revenue, while the top 60% contributed to more than 80%, a strong concentration of revenue among higher-value segments.

In addition, 41.4% of customers were identified as non-spenders, representing an opportunity for increased customer engagement and conversion to spending customers. Geographic analysis revealed regional revenue concentration in Washington state, U.S., which accounted for 46.6% of total online sales revenue. High-performing cities were also found to be clustered in Washington. This makes sense given that Funky's is headquartered in Bellingham, Washington, explaining stronger brand recognition and local customer support, especially with appearances in local markets. Finally, a statistically significant positive relationship between the proportion of subscribed customers and customer spending at the city level further supports the conclusion that marketing engagement is associated with increased customer spending.

Recommendations

Based on the findings of this analysis, strategic recommendations can be made to support business growth. First, focus on marketing subscription efforts should be prioritized. Increasing email and/or SMS marketing acceptance rates and offering targeted incentives to current subscribers is likely to improve customer spending and store engagement. This can be achieved through discounting new online customers (and promoting this through various media outlets), targeted promotions through social media marketing, and incentives designed to encourage subscription.

Second, targeted strategies should be developed to convert non-spending customers into active spenders. Personalized outreach and first-purchase incentives would be particularly effective in this area.

Third, Funky's should also focus on strengthening relationships with high-value customers by introducing loyalty programs or promotions aimed at encouraging repeat purchases such as bundles. These customers already contribute a significant share of online sales revenue and represent the opportunity for sustainable growth.

Fourth, efforts should be made to expand marketing initiatives across North America. Targeting underperforming regions with tailored campaigns may introduce new growth opportunities and increase online sales revenue in different regional markets.

Lastly, social media networks are vital in the popularization of products in today's society. Sending public relations (PR) packages with Funky's Hot Sauce Factory products and/or partnering with social influencers online could help introduce new customers to Funky's. The brand's hot sauce products have consistently been featured on Hot Ones, a YouTube

talk show series that hosts celebrities who answer questions while eating chicken wings with various hot sauces (Hot Ones). More product features as such would likely increase brand visibility, attract customers, and strengthen customer engagement. Continued use of data-driven insights will be essential in guiding these strategies and optimizing future decision-making.

Conclusion

This analysis of customer behavior at Funky's Hot Sauce Factory provides insight into the KPIs that influence spending and online sales revenue generation. The findings demonstrate customer spending distribution and concentration, the association between marketing engagement and spending, and geographic sales patterns.

Overall, these results suggest that strategic focus on targeted marketing engagement, non-spending customer conversion, and geographic expansion may support continued business growth. By leveraging data-driven insights to better understand customer behavior, Funky's Hot Sauce Factory is well-positioned to improve customer relationships and and identify new opportunities for business expansion.

References

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